Leveraging Kinematic space for Deep Representation Learning

Adarsh Jamadandi
Computer Vision and Graphics Lab
KLE Technological University
Hubli, India
adarsh.jamadandi@kletech.ac.in

Uma Mudenagudi
Computer Vision and Graphics Lab
KLE Technological University
Hubli, India
uma@kletech.ac.in

Abstract

Learning hierarchical representations in curved Riemannian manifolds has gained traction. More specifically, models of hyperbolic geometry have been used for learning symbolic data with hierarchical structure. While, embedding spaces with non-constant sectional curvature offer potentially more powerful representation capacity, their utility in machine learning pipelines is primarily limited by the in-extensibility of Riemannian optimization tools. In this work, we investigate Kinematic space, an intermediary space of bulk geodesics local to integral geometry and its utility in acting as geometrical inductive biases. Our formulation includes adopting the Poincaré upper half-plane model, which is computationally intractable and so far has not found utility in machine learning pipelines. We leverage Kinematic space to transform geometrical information such as geodesic distance from upper half-plane model to a maximally symmetric pseudo-Riemannian manifold called de Sitter space, where Riemannian optimization methods are applicable. We provide a principled approach to train a Variational Autoencoder (VAE) whose latent space is the de Sitter space. Extensive experiments suggest consistent improvements over existing hyperbolic and Euclidean versions of VAE.

1 Introduction

It is hypothesized that the quality of embeddings largely depends on the embedding space and how well it captures data structure. This realization has led to the resurgence of the Manifold hypothesis [3], which asserts that raw data (like pixels of images) lie on a low-dimensional manifold but with a new imperative - explicit assumption of the underlying manifold, like hyperbolic spaces [31, 32] and spherical spaces [13, 29]. Hierarchical data benefits from hyperbolic space; for example, datasets with properties like hierarchy and scale-free can be embedded with smaller distortions in hyperbolic geometry. This realization has manifest hyperbolic geometry models such as Poincaré ball and Hyperboloid as the new workhorse, for a variety of applications such as Word embeddings [31, 32], hyperbolic versions of Neural Networks [15] and Graph Neural Networks [7, 33] and even to model complex networks such as Protein-Protein interaction network [1, 23].

The idea has been successfully extended to Generative models, such as Variational Autoencoders (VAEs) as well. The latent space is modified to be hyperbolic [30, 28] and spherical [13] to account for the datasets' anatomy. In this work, we introduce a powerful transformation geometry called Kinematic space, used in integral geometry [35] and theoretical physics [10, 39] to obtain computationally tractable pseudo-Riemannian Lorentzian manifolds which can be used in variety of representation learning tasks as geometrical inductive biases. We choose our embedding space to be the Poincare upper-half-plane model ($\mathbb{U}_p$), which is rarely considered in the literature due to its intractability. By using Kinematic space, we effectively re-interpret metrics (such as geodesic

NewinML Workshop, 34th Neural Information Processing Systems (NeurIPS 2020), Vancouver, Canada.

distance and exponential map) into a more computationally tractable manifold called de Sitter space ($de\mathbb{S}_2$) [9], which is a Lorentzian manifold, where the application of Riemannian optimization tools is restored. We generalize the notion of VAEs to pseudo-Riemannian manifolds and report consistent improvements over existing hyperbolic and Euclidean versions. Our contributions can be highlighted as follows :

1. We introduce kinematic space, an auxiliary Lorentzian geometry as a powerful transformation tool that can re-interpret geometrical information such as geodesic distance and exponential maps from one geometrical space to another. Leveraging this formulation, we show that learning representations in the upper half-plane model is equivalent to learning in a maximally symmetric pseudo-Riemannian manifold called de Sitter space, where Riemannian optimization methods are applicable.

2. We formulate Wrapped Normal Distribution in kinematic Space and use it to generalize the notion of Variational Autoencoders, which we call $\mathcal{K}_s-$VAE, with Gaussian-like priors constructed in this newly proposed space. We perform extensive experiments on different standard benchmark datasets and show consistent improvements over existing Euclidean and Hyperbolic versions.

2 Related Work

In this section, we discuss a few contemporary works that have adopted the non-Euclidean realm as their workhorse for a variety of applications. Authors in [31, 32] spearheaded the idea of using Poincaré and Lorentz models of hyperbolic geometry with deep neural networks for learning parsimonious representations of symbolic datasets. With the advent of optimization techniques for Riemannian manifolds [5, 2], the idea of using hyperbolic geometry in various settings has gained traction, for example, Hyperbolic Neural Networks [15], Hyperbolic Attention Mechanisms for machine translation [18]. Authors in [17] propose product spaces of mixed-curvature manifolds for learning representations. Results from Network Science also advocates the existence of hyperbolic geometry as the latent space for many applications, for example, authors in [23], show that the properties of complex networks such as heterogeneous degree distributions and strong clustering are reflections of the negative curvature of latent hyperbolic geometry underlying the complex networks.

To facilitate solving of probabilistic inference problems with an inherent hierarchical structure, authors in [30] introduce a pseudo-Hyperbolic Gaussian distribution and demonstrate probabilistic word embeddings and hyperbolic version of Variational Autoencoders (HVAE). Graph Neural Networks have been at the helm of graph representation learning 38, 19, 22, which embed the graph nodes in Euclidean space. This introduces distortions, which can be mitigated by Hyperbolic versions of Graph Neural Networks[7, 33]. Hyperbolic space has also proliferated in areas like Latent Variable Modeling in hyperbolic space [6] and Mixed-curvature VAEs [36].

Motivated by these lines of work, we introduce a novel geometrical inductive - kinematic space, a pseudo-Riemannian Lorentzian space to learn representations. The utility of kinematic space has been extensively found in the areas of theoretical physics [12, 39, 11]. Leveraging ideas from different sciences is non-trivial, and to our knowledge, this is the first work that introduces kinematic space in the context deep representation learning.

3 Review of Differential Geometry

In this section, we provide a quick primer on basic tools used in differential geometry and enumerate the different hyperbolic geometry models commonly used in literature as embedding space.

1. **Riemannian Manifolds -**A Riemannian manifold $(\mathcal{M}, g)$ is a real, smooth manifold equipped with an inner product g_p on the tangent space $\mathcal{T}_p\mathcal{M}$ at each point $p \in \mathcal{M}$ given by, $g_p : \mathcal{T}_p\mathcal{M} \times \mathcal{T}_p\mathcal{M} \to \mathbb{R}$. The family of inner products g_p is called the Riemannian metric tensor used to define geometrical concepts like length and angles on the Riemannian manifold.

2. **Parallel Transport -** Let $\mathcal{C}(\lambda)$ be a curve parametrized by λ with tangent $\vec{V}$ and a vector $\vec{a}$ defined at some point on the curve $\mathcal{C}$. Parallel transport involves defining this vector $\vec{a}$ along each point of the curve such that,

$$\frac{d\vec{a}}{d\lambda} = 0 \quad \text{and} \quad \nabla_V \vec{a} = 0 \quad \text{where, } \nabla_V \text{is the covariant derivative along V.} \tag{1}$$

Simply put, parallel transport $PT_{\mathbf{o}\to\mathbf{u}}$ maps a vector $p \in \mathcal{T}_o\mathcal{M}$ to $PT_{\mathbf{o}\to\mathbf{u}}(p) \in \mathcal{T}_u\mathcal{M}$

3. **Geodesics -** Geodesics γ are generalizations of straight lines in Euclidean space to curved manifolds. In terms of Parallel transport, we can define geodesics as special type of curves that parallel transports its own tangent vector (i.e $\nabla_V \vec{V} = 0$).

3.1 Models of Hyperbolic Geometry

1. **Poincaré Ball Model :** is a Riemannian manifold $(\mathcal{B}, g_B)$, where $\mathcal{B} = \{x \in \mathbb{R}^n : ||x|| < 1\}$ is an open unit ball. The distance between points $x, y \in \mathcal{B}$ is given by,

$$d_{\mathcal{B}}(x, y) = \operatorname{arcosh}\left(1 + 2\frac{||x-y||^2}{(1-||x||^2)(1-||y||^2)}\right) \tag{2}$$

2. **Lorentz Model :** also referred to as the Hyperboloid model, is a Riemannian manifold $(\mathcal{L}, g_L)$, where $\mathcal{L} = \{\mathbf{x} \in \mathbb{R}^{n+1} : \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -1, x_o > 0\}$ and the Minkowski inner product defined by,

$$\langle x, y \rangle_{\mathcal{L}} := -x_0 y_0 + x_1 y_1 + ... + x_n y_n \tag{3}$$

The induced distance function is given by,

$$d_{\mathcal{L}}(\mathbf{x}, \mathbf{y}) = \operatorname{arcosh}(-\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}) \tag{4}$$

The Lorentz model avoids numerical instabilities [32] and is usually the preferred geometrical inductive.

3. **Poincaré upper Half-Plane Model:** is a Riemannian manifold $\mathbb{U}_P = \{(\mathbf{x}, \mathbf{y} \in \mathbb{R}), \mathbf{y} > 0\}$, its distance metric is given by,

$$d_{\mathbb{H}_{UP}}(\mathbf{x}, \mathbf{y}) = ((x_1, y_1), (x_2, y_2)) = \operatorname{arcosh}\left(1 + \frac{(x_2 - x_1)^2 + (y_2 - y_1)^2}{2y_1 y_2}\right) \tag{5}$$

4 Kinematic space

In this section, we outline the mathematical framework that underlies the idea of kinematic space. Our treatment of kinematic space ($\mathcal{K}_s$), is inspired by [35, 10]. Suppose we are interested in measuring the length of a curve $\mathcal{C}$ on an Euclidean plane, we can draw a number of tangents $\mathbf{t}$ to the curve. The equation of the straight line given by

$$x \cos\theta + y \sin\theta - l = 0 \tag{6}$$

where θ is the polar angle and l is the distance of straight line from the origin as shown in Figure 1 A. We can estimate the length of the curve $\mathcal{C}$ by the Crofton's formula [35],

$$\text{Length} = \frac{1}{4}\int_0^{2\pi} d\theta \int_{-\infty}^{+\infty} \eta(\theta, l) dl \tag{7}$$

where $\eta(\theta, l)$ gives the intersection number of the tangents and the curve. This space of oriented geodesics panning $\theta \in [0, 2\pi]$ and $l \in (-\infty, \infty)$ is called the *kinematic space of the Euclidean plane*. Carrying out the same formulation on the hyperbolic plane ($\mathbb{H}^n$), the straight lines are now replaced by geodesics given by equation

$$\tanh\rho \cos(\hat{\theta} - \theta) = \cos\alpha \tag{8}$$

where, α is the opening angle of the geodesic and θ is the angular coordinate of the center of the geodesic. The kinematic space of the hyperbolic plane is now the space of geodesics panned by

$\theta \in [0, 2\pi]$ and $\alpha \in [0, \pi]$. This is represented in Figure 1 B, where a geodesic γ is drawn on the Poincaré upper half-plane model ($\mathbb{U}_P$). Its stereographic projection is shown on the Poincaré disk ($\mathbb{P}_d$). Thus, kinematic space is an auxiliary Lorentzian space of oriented geodesics that acts as an intermediary for translating information from the space of length of curves to a space where the length of the curves can be expressed as volume of intersecting lines. In Figure 1C, we see a visualization of curves $\mathbf{C}_i = \{\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4\} \in \mathbb{H}^n$ and their corresponding kinematic space $\mathcal{K}_s = \{\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3, \mathbf{K}_4\} \in \{\theta \in [0, 2\pi], \alpha \in [0, \pi]\}$.

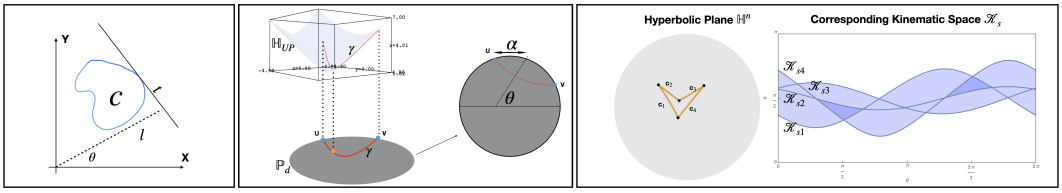

Figure 1A Figure 1B Figure 1C

Figure 1: A)The Crofton's formula can be used to estimate the length of a closed curve $\mathcal{C}$. Parametrized by (θ, l), we can express the length of the curve as a volume of straight lines intersecting it. B) A geodesic γ drawn on the upper half-plane model can be stereographically projected on to the Poincaré disk, parameterized by the opening angle α and the angular coordinate θ. This space of oriented geodesics is the kinematic space of the Hyperbolic Plane. Curves drawn on the hyperbolic plane and their corresponding kinematic space $\in \{\theta \in [0, 2\pi], \alpha \in [0, \pi]\}$ is shown in Figure 1C.

4.1 Why Kinematic space?

Hyperbolic geometry models have become de-facto workhorse for modeling hierarchical data. However, many real-world graphs also exhibit properties such as *Causality*, (for example, citation networks), which is not captured by hyperbolic geometry. As authors in [32] describe, tasks like - inferring concept hierarchies in large taxonomies of data involves defining a *partial order set* (Posets) over entities, and the chosen embedding space should be capable of preserving this order. These partial order sets are also called as *Causal Sets* in theoretical physics and are directly related to Lorentzian Manifolds [24]. Thus, the topology of structural datasets with hierarchical anatomy and/or expressing causal relationships are more naturally described by Lorentzian manifolds. To substantiate this, we generate a random directed acyclic graph (DAG), with high $\delta-$hyperbolicity = 6.5. The concept of hyperbolicity (see B) introduced by Gromov [16], measures the *tree-likeliness* of the data. Tree-like ($\delta = 0$) nature allows a data to be isometrically embedded into hyperbolic space. We use Multi-dimensional scaling algorithm [8, 34] to embed the graph both in 2- dimensional de Sitter space and Poincaré ball model. We measure the area under curve (AUC) to assess the quality of embedding. As evident from the first row of Figure 2, embedding the graph in de Sitter space yields 55% accuracy while trying to embed in hyperbolic space fails miserably.

4.2 de Sitter space ($de\mathbb{S}_2$)

The de Sitter space is a maximally symmetric, Lorentzian manifold with constant positive curvature. By using the formulation outlined in Section 4 we show that the kinematic space of the upper half-plane model is the de Sitter space, and use it for learning representations.

Proposition 4.1.1 Let $de\mathbb{S}_2$ be the $(d+1)$ dimensional de Sitter space in the $(d+2)$ dimensional Minkowski space $\mathbb{M}$ visualized as a single sheeted hyperboloid with pseudo-radius λ given by $-z_0^2 + z_1^2 + z_3^2 + ... + z_n^2 = \lambda^2 = \frac{1}{K}$. The metric is induced by the embedding Minkowski space [9], and the distance function is given by

$$d_{de\mathbb{S}_2}(\mathbf{x}, \mathbf{y}) = \lambda \operatorname{arcosh}\left(\frac{-\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}}{\lambda^2}\right) \tag{9}$$

The exponential map $\exp_p : \mathcal{T}_p de\mathbb{S}_2 \to de\mathbb{S}_2$ is

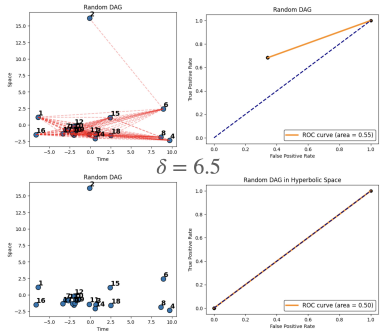

Figure 2: For a random Directed Acyclic Graph with $\delta = 6.5$, we try to embed in both de Sitter space (Top Row) and hyperbolic space (Bottom Row). From the figure, its evident that, graphs, especially DAGs are more naturally embedded in Lorentzian manifolds.

$$\exp_p(\mathbf{v}) = \cosh(\sqrt{K}||\mathbf{v}||_{\mathcal{L}})\mathbf{p} + \mathbf{v}\frac{\sinh(\sqrt{K}||\mathbf{v}||_{\mathcal{L}})}{\sqrt{K}||\mathbf{v}||_{\mathcal{L}}} \tag{10}$$

The logarithmic map $\log_p : \mathcal{T}_p de\mathbb{S}_2 \to de\mathbb{S}_2$ is

$$\log_p(\mathbf{y}) = \frac{\operatorname{arcosh}(-K\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}})}{\sinh(\operatorname{arcosh}(-K\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}}))}(\mathbf{y} - K\langle \mathbf{x}, \mathbf{y}\rangle_{\mathcal{L}}\mathbf{p}) \tag{11}$$

Proposition 4.1.2 $\mathbb{U}_P \rightleftharpoons_{\mathcal{K}_s} de\mathbb{S}_2$ The kinematic space ($\mathcal{K}_s$) of the upper half-plane model ($\mathbb{U}_P$) is the $(d+1)$ dimensional de Sitter space ($de\mathbb{S}_2$) visualized as a single sheeted hyperboloid in the Minkowski space $\mathbb{M}^{d+2}$, and there is a canonical identification between the geodesics $\gamma \in \mathbb{U}_P$ and the points in $\mathcal{K}_s$.

Proposition 4.1.3 For every geodesic γ that can be drawn on $\mathbb{U}_P$, we can find a unique plane $\mathbf{p}$ intersecting $\mathbb{U}_P$, whose normal $\mathbf{n}$ at the origin corresponds to a point in $de\mathbb{S}_2$. This is shown in Figure 3. The proof showing the correspondence between $\mathbb{U}_P$ and $de\mathbb{S}_2$ is available in the Appendix.

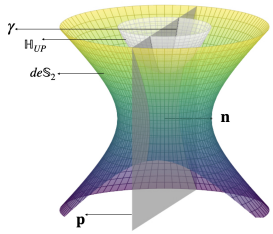

Figure 3: The kinematic space $\mathcal{K}_s$ of the upper half-plane model $\mathbb{U}_P$ is the de Sitter space $de\mathbb{S}_2$ visualized as a single-sheeted hyperboloid with its distance metric induced by the ambient Minkowski space.

5 Constructing Gaussian-like distribution in de Sitter space

We present a Gaussian-like distribution on de Sitter space whose density can be investigated analytically and is differentiable with respect to its parameters. This formulation has been successfully adapted to hyperbolic space [30] to facilitate solving probabilistic inference problems in hyperbolic space to tackle datasets with inherent hierarchical anatomy. To demonstrate the efficacy of our proposed geometrical inductive we introduce $\mathcal{K}_s$-VAE, a generalization of VAEs to de Sitter space and report improvements over existing Euclidean [37] and Hyperbolic versions[30].

To define Gaussian-like distribution on the tangent space, we first start by describing the distribution at the origin of our prescribed geometry, and using a combination of exponential map and parallel

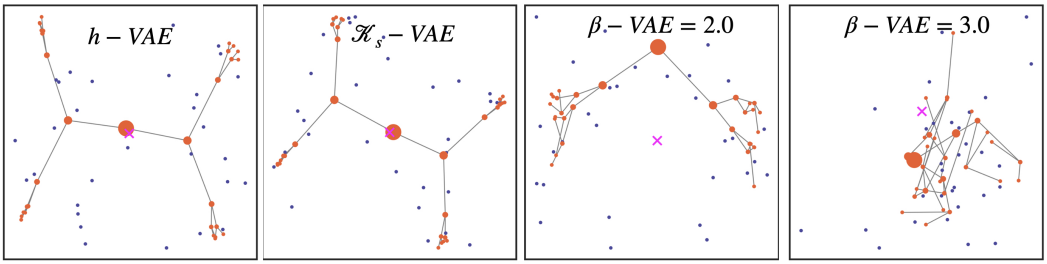

Figure 4: The hyperbolic-VAE [30] and $\mathcal{K}_s$-VAE automatically discover the hierarchical nature of the synthetic tree dataset. The Vanilla-VAE with $\beta = \{2, 3\}$ is shown in the third and fourth figure respectively. The method fails to account for the hierarchical structure of the dataset, leading to poor correlation with hamming distance.

transport, we move the distribution at any desired location in our space. The steps for constructing pseudo-Hyperbolic Gaussian, $\mathcal{G}_{de\mathbb{S}_2}(\mu, \Sigma)$, where $\mu \in \text{de}\mathbb{S}_2$, can be summarized as follows :

1. Sampling a vector $\mathbf{v}$ from the Gaussian distribution $\mathcal{N}(0, \Sigma)$ defined over $\mathbb{R}^n$.
2. Parallel transporting $\mathbf{v}$ from the tangent space $\mathbf{o}$ to the tangent space of new point $\mathbf{u}$ to obtain $\mathbf{j}$ by using the formula,

$$PT_{\mathbf{o}\to\mathbf{u}}(\mathbf{v}) = \mathbf{v} + \frac{K\langle y, u\rangle_{\mathcal{L}}}{1 + K\langle o, u\rangle_{\mathcal{L}}}(o + u) \tag{12}$$

3. Map the point $\mathbf{j}$ to the manifold using the exponential map at $\mathbf{u}$ given by the Equation 10 .

To calculate the probability density of $\mathcal{G}_{de\mathbb{S}_2}(\mu, \Sigma)$ [6], we can use a combination of parallel transport (Equation 12) and exponential map (Equation 10), to obtain,

$$\log g(\mathbf{z}) = \log g(v) - (n-1)\log\left(\frac{\sinh ||\mathbf{j}||}{||\mathbf{j}||}\right) \tag{13}$$

where, $\log g(\mathbf{z})$ is the wrapped normal distribution and $\log g(v)$ is the normal distribution in tangent space of $\mathbf{o}$.

5.1 Variational Auto Encoders in Kinematic Space

We adopt a strategy similar to authors in [30] and formulate a Kinematic Space version of VAE ($\mathcal{K}_s$-VAE) where the latent variables are described in de$\mathbb{S}_2$. The formulation of a VAE involves maximizing the sum of evidence lower bound (ELBO) defined for a data point $x_\mathbf{i} \in \mathcal{D} = \{x_\mathbf{i}\}_{i=1}^N$, by training an encoder model $\varepsilon_\psi(z|x)$ and decoder model $\mathbf{d}_\theta(x|z)$ jointly given by,

$$\log d_\theta(x^{(\mathbf{i})}) \geq \mathbb{L}(\theta, \psi; x^{(\mathbf{i})}) = \mathbb{E}_{\varepsilon_\psi(z|x^{(\mathbf{i})})}\left[\log d_\theta(x^{(\mathbf{i})}|z)\right] - \text{KL}_{\text{Div}}(\varepsilon_\psi(z|x^{(\mathbf{i})})||d_\theta(z)) \tag{14}$$

Here, $\varepsilon_\psi(z|x_\mathbf{i})$ is the variational posterior distribution. The kinematic space version modifies the classic VAE by defining $\varepsilon_\psi = \mathcal{G}_{de\mathbb{S}_2}(\mu, \Sigma)$. To ensure we get the output of the encoder $\varepsilon_{out} \in \text{de}\mathbb{S}_2$, we apply the exponential map 10 . A concurrent work by authors [36] also propose a Gaussian-like prior and formulate VAE in a mixed curvature product space with fixed and trainable curvature, the main difference with our work is that our focus here is on investigation of kinematic space as an effective geometric space for learning deep representations and we demonstrate this by formulating $\mathcal{K}_s$-VAE as an application.

6 Experiments

For our experiments we use two datasets as enumerated below. Synthetic Binary Tree - We generate a binary tree dataset of depth 5. We followed the method proposed by authors in [30]. A binary

Table 1: We report log-likelihood for different latent dimensions. Our method outperforms [20] and [30] for all dimensions. We see an improvement in the log-likelihood values. For instance, for $n = 20$ we see an increase of 21.75% in log-likelihood.

Dimension	β-VAE[20] (LL)	h-VAE[30](LL)	**Ours**(LL)
2	-143.16 $\pm$ 2.72	-145.61 $\pm$0.45	**-140.88** $\pm$ **4.04**
5	-105.78$\pm$40.51	-109.22$\pm$0.61	**-106.09** $\pm$ **0.72**
10	-86.25$\pm$0.52	86.40$\pm$0.28	**-79.44** $\pm$ **0.62**
20	-77.89$\pm$0.36	-79.23$\pm$0.20	**-61.99**$\pm$**0.18**

Table 2: We measure the correlation of hamming distance of the nodes in the tree and in the latent space in presence and absence of noise. The Vanilla-VAE with different β fails to account for the hierarchical structure of the dataset. Our method outperforms both β-VAE [20] and hyperbolic-VAE [30], indicating that the proposed geometrical space is better inductive.

	Model	Correlation	Correlation with Noise
Vanilla [20]	β=1.0	0.86 $\pm$ 0.06	0.70 $\pm$ 0.00
	β=2.0	0.74 $\pm$ 0.00	0.59 $\pm$ 0.04
	β=3.0	0.35 $\pm$ 0.09	0.06 $\pm$ 0.00
	h-VAE [30]	0.87 $\pm$ 0.00	0.64 $\pm$ 0.00
	Ours	**0.89**$\pm$ **0.00**	**0.72** $\pm$ **0.05**

tree representation is obtained such that the hamming distance in the original graph correlates to the distance between the nodes in the binary representation. MNIST - We use a binarized version of MNIST for $\mathcal{K}_s-$ VAE as described by authors in [30]. We measure the reconstruction performance quantified by log-likelihood.

7 Results and Discussion

Synthetic Binary Tree We use a MLP of 100 hidden units and depth 3. The learning rate is 0.001 and we train for 1200 iterations and 255 batch size. We measure the correlation of the hamming distance between the nodes of the synthetic tree and in the latent space. The red points in Figure 4 show the actual data points while the blue points indicate the presence of noisy points. From the figure, its evident that $\beta-$VAE fails to preserve the hierarchical structure of the dataset. The hyperbolic-VAE and our method is successful in automatically discovering the hierarchical nature. This is due to the appropriate geometrical inductive bias that we have equipped our VAEs with. Note a slight offset in the pink $\times$, in the hyperbolic-VAE, while the $\times$ coincides perfectly in case of $\mathcal{K}_s-$VAE, indicating de Sitter space is a better alternative to models of hyperbolic geometry. The corresponding quantitative results are summarized in Table 2. Our proposed framework fairs better even in the presence of noise.

Binarized MNIST We use a MLP of 500 hidden units and depth 3. The number of iterations are 150000 with batch size 100. We measure the ability of $\mathcal{K}_s-$VAE to reconstruct the MNIST dataset [25]. The performance is quantified by log likelihood. We compare with $\beta-$VAE [20] and hyperbolic-VAE [30]. The results are summarized in Table 1. For varying latent dimensions, we see our method outperforms both the versions. All the experiments were averaged over 2 trials. Please refer to supplementary material for additional results.

8 Conclusion

In this paper, we have introduced a novel Lorentzian geometry called kinematic space that helps transform geometric information such as geodesic distance from one space to another by casting it as an integral geometric problem. We have provided a principled way for training VAEs whose latent space is the de Sitter space, a pseudo-Riemannian manifold. Extensive experiments on various datasets show the utility of Lorentzian manifolds in machine learning pipelines.

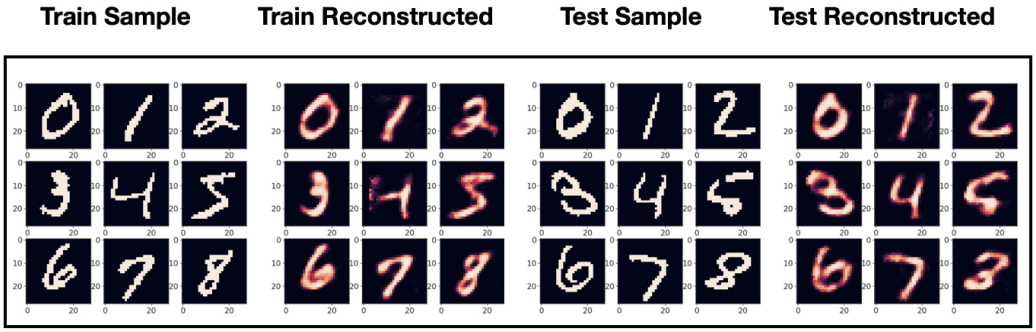

Figure 5: Visualization of binarized MNIST reconstruction for $n = 20$.

References

[1] G. Alanis-Lobato, P. Mier, and M. A. Andrade-Navarro. The latent geometry of the human protein interaction network. *bioRxiv*, 2017. doi: 10.1101/213165. URL `https://www.biorxiv.org/content/early/2017/11/02/213165`.

[2] G. Becigneul and O.-E. Ganea. Riemannian adaptive optimization methods. In *International Conference on Learning Representations*, 2019. URL `https://openreview.net/forum?id=r1eiqi09K7`.

[3] Y. Bengio, A. Courville, and P. Vincent. Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8):1798–1828, Aug 2013. ISSN 0162-8828. doi: 10.1109/TPAMI.2013.50.

[4] G. Bianconi and C. Rahmede. Emergent hyperbolic network geometry. *Scientific Reports*, 7(1): 41974, 2017. doi: 10.1038/srep41974. URL `https://doi.org/10.1038/srep41974`.

[5] S. Bonnabel. Stochastic gradient descent on riemannian manifolds. *IEEE Transactions on Automatic Control*, 58(9):2217–2229, Sep. 2013. doi: 10.1109/TAC.2013.2254619.

[6] A. J. Bose, A. Smofsky, R. Liao, P. Panangaden, and W. L. Hamilton. Latent variable modelling with hyperbolic normalizing flows. *Proceedings of the 37th International Conference on Machine Learning*, 2020.

[7] R. Y. C. R. J. L. Chami, I. Hyperbolic graph convolutional neural networks. In *Neural Information Processing Systems, NeurIPS*, 2019.

[8] J. R. Clough and T. S. Evans. Embedding graphs in lorentzian spacetime. *PLOS ONE*, 12 (11):1–14, 11 2017. doi: 10.1371/journal.pone.0187301. URL `https://doi.org/10.1371/journal.pone.0187301`.

[9] W. J. Cunningham, D. Rideout, J. Halverson, and D. Krioukov. Exact geodesic distances in flrw spacetimes. *Phys. Rev. D*, 96:103538, Nov 2017. doi: 10.1103/PhysRevD.96.103538. URL `https://link.aps.org/doi/10.1103/PhysRevD.96.103538`.

[10] B. Czech, L. Lamprou, S. McCandlish, and J. Sully. Integral Geometry and Holography. *JHEP*, 2015. doi: 10.1007/JHEP10(2015)175.

[11] B. Czech, L. Lamprou, S. McCandlish, B. Mosk, and J. Sully. A stereoscopic look into the bulk. *Journal of High Energy Physics*, 2016(7):129, 2016. doi: 10.1007/JHEP07(2016)129. URL `https://doi.org/10.1007/JHEP07(2016)129`.

[12] B. Czech, L. Lamprou, S. McCandlish, and J. Sully. Tensor networks from kinematic space. *Journal of High Energy Physics*, 2016(7):100, 2016. doi: 10.1007/JHEP07(2016)100. URL `https://doi.org/10.1007/JHEP07(2016)100`.

[13] T. R. Davidson, L. Falorsi, N. De Cao, T. Kipf, and J. M. Tomczak. Hyperspherical variational auto-encoders. *34th Conference on Uncertainty in Artificial Intelligence (UAI-18)*, 2018.

[14] A. Finke and A. H. Thiery. On importance-weighted autoencoders, 2020. URL https://openreview.net/forum?id=ryg7jhEtPB.

[15] O.-E. Ganea, G. Bécigneul, and T. Hofmann. Hyperbolic neural networks. In *NeurIPS*, pages 5350–5360, 2018. URL http://papers.nips.cc/paper/7780-hyperbolic-neural-networks.

[16] M. Gromov. *Hyperbolic Groups*.

[17] A. Gu, F. Sala, B. Gunel, and C. Ré. Learning mixed-curvature representations in product spaces. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=HJxeWnCcF7.

[18] C. Gulcehre, M. Denil, M. Malinowski, A. Razavi, R. Pascanu, K. M. Hermann, P. Battaglia, V. Bapst, D. Raposo, A. Santoro, and N. de Freitas. Hyperbolic attention networks. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=rJxHsjRqFQ.

[19] W. Hamilton, Z. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems 30*. 2017.

[20] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. M. Botvinick, S. Mohamed, and A. Lerchner. beta-vae: Learning basic visual concepts with a constrained variational framework. In *ICLR*, 2017.

[21] A. Jamadandi, R. Tabib, and U. Mudenagudi. Learning hierarchical representations in kinematic space. In *Graph Representation Learning Workshop, NeurIPS*, 2019.

[22] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*. 2017.

[23] D. Krioukov, F. Papadopoulos, M. Kitsak, A. Vahdat, and M. Boguñá. Hyperbolic geometry of complex networks. *Phys. Rev. E*, 82:036106, Sep 2010. doi: 10.1103/PhysRevE.82.036106. URL https://link.aps.org/doi/10.1103/PhysRevE.82.036106.

[24] D. Krioukov, M. Kitsak, R. S. Sinkovits, D. Rideout, D. Meyer, and M. Boguñá. Network cosmology. *Scientific Reports*, 2(1):793, 2012. doi: 10.1038/srep00793. URL https://doi.org/10.1038/srep00793.

[25] Y. LeCun. The mnist database of handwritten digits. 1998. URL http://yann.lecun.com/exdb/mnist/.

[26] M. Leimeister and B. J. Wilson. Skip-gram word embeddings in hyperbolic space. *CoRR*, abs/1809.01498, 2018. URL http://arxiv.org/abs/1809.01498.

[27] A. Lou, I. Katsman, Q. Jiang, S. Belongie, S.-N. Lim, and C. D. Sa. Differentiating through the fréchet mean, 2020.

[28] E. Mathieu, C. Le Lan, C. J. Maddison, R. Tomioka, and Y. W. Teh. Continuous hierarchical representations with poincaré variational auto-encoders. In *Advances in Neural Information Processing Systems 32*, pages 12565–12576. Curran Associates, Inc., 2019.

[29] Y. Meng, J. Huang, G. Wang, C. Zhang, H. Zhuang, L. Kaplan, and J. Han. Spherical text embedding, 2019.

[30] Y. Nagano, S. Yamaguchi, Y. Fujita, and M. Koyama. A wrapped normal distribution on hyperbolic space for gradient-based learning. In *ICML*, 2019.

[31] M. Nickel and D. Kiela. Poincaré embeddings for learning hierarchical representations. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems 30*, pages 6338–6347. Curran Associates, Inc., 2017. URL http://papers.nips.cc/paper/7213-poincare-embeddings-for-learning-hierarchical-representations.pdf.

[32] M. Nickel and D. Kiela. Learning continuous hierarchies in the lorentz model of hyperbolic geometry. *CoRR*, abs/1806.03417, 2018. URL `http://arxiv.org/abs/1806.03417`.

[33] M. N. Qi Liu and D. Kiela. Hyperbolic graph neural networks. In *Neural Information Processing Systems, NeurIPS*, 2019.

[34] C. D. Sa, A. Gu, C. Ré, and F. Sala. Representation tradeoffs for hyperbolic embeddings, 2018.

[35] L. A. Santaló and M. Kac. *Integral Geometry and Geometric Probability*. Cambridge University Press, 2004.

[36] O. Skopek, O.-E. Ganea, and G. Bécigneul. Mixed-curvature variational autoencoders. In *International Conference on Learning Representations*, 2020. URL `https://openreview.net/forum?id=S1g6xeSKDS`.

[37] L. Vilnis and A. McCallum. Word representations via gaussian embedding. In *International Conference on Learning Representations*, 2015.

[38] K. Xu, W. Hu, J. Leskovec, and S. Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations*. 2019.

[39] J.-d. "Zhang and B. Chen. Kinematic space and wormholes. *Journal of High Energy Physics*, 2017. doi: 10.1007/JHEP01(2017)092. URL `https://doi.org/10.1007/JHEP01(2017)092`.

A Fundamental Connection between $de\mathbb{S}_2$ and Upper half-plane model

In this section we provide proof showing the connection between de Sitter space (de$\mathbb{S}_2$) and upper half-plane model($\mathbb{U}_P$). Our treatment is similar to that of authors in [39].

Lemma A.1. *Proof for showing the kinematic space of Upper half plane model is de Sitter space. The kinematic space ($\mathcal{K}_s$) of the Upper half-plane model ($\mathbb{U}_P$) is the $(d+1)$ dimensional de Sitter space ($de\mathbb{S}_2$) visualized as a single sheeted hyperboloid in the Minkowski space $\mathbb{M}^{2+1}$, and there is a canonical identification between the geodesics $\gamma \in \mathbb{U}_P$ and points in $\mathcal{K}_s$.*

Proof. Proving this proposition involves showing that for every geodesic $\gamma \in \mathbb{U}_P$ theres a unique plane intersecting $\mathbb{U}_P$ and the normal vector from the plane gives rise to a pair of anti-podal points on $de\mathbb{S}_2$ which represent that geodesic.

We start by embedding $\mathbb{U}_P$ and $de\mathbb{S}_2$ in the same 3-d flat space $\mathbb{R}^{2,1}$ -

The $\mathbb{U}_P$ is given by,

$$X^2 - Y^2 - Z^2 = 1 \tag{15}$$

By introducing coordinates : $X = \cosh\rho$, $Y = \sinh\rho\cos\theta$ and $Z = \sinh\rho\sin\theta$, we can recover the metric of $\mathbb{U}_P$.

$$ds^2 = -dX^2 + dY^2 + dZ^2 \tag{16}$$

Similarly for $de\mathbb{S}_2$,

$$-X^2 + Y^2 + Z^2 = 1 \tag{17}$$

we can introduce coordinates : $Y = \cosh\tau\cos\theta$, $Z = \cosh\tau\sin\theta$ and $X = \sinh\tau$, to get the metric of $de\mathbb{S}_2$,

$$ds^2 = -d\tau^2 + \cosh^2\tau d\theta^2 \tag{18}$$

By introducing another transformation for ($\alpha \in (0, \pi)$),

$$\cosh\tau = \frac{1}{\sinh\alpha} \tag{19}$$

Then Equation 17 becomes,

$$ds^2 = \frac{-d\alpha^2 + d\theta^2}{\sin^2 \alpha} \tag{20}$$

A point (θ, α) will correspond to $\left(-\cot\alpha, \frac{\cos\theta}{\sin\alpha}, \frac{\sin\theta}{\sin\alpha}\right)$ in the coordinates of (X, Y, Z).

The equation for geodesics in $\mathbb{U}_P$ is given by,

$$\tanh\rho\cos(\theta - \theta_0) = \cos\alpha_0 \tag{21}$$

The same equation for coordinates in $\mathbb{R}^{2,1}$ is given by,

$$\cos\alpha_0 X - \cos\theta_0 Y - \sin\theta_0 Z = 0 \tag{22}$$

which is the equation of plane, thus for every geodesic γ that can be drawn on $\mathbb{U}_P$, we can find a unique plane $\mathbf{p}$ intersecting $\mathbb{U}_P$, whose normal $\mathbf{n}$ at the origin corresponds to a point in $de\mathbb{S}_2$. This is shown in Figure 3. The coordinates of these points are $\mp\left(\cot\alpha, \frac{\cos\theta}{\sin\alpha}, \frac{\sin\theta}{\sin\alpha}\right)$ which is similar to the coordinates given by Equation 20. The $\mp$ indicates the orientation of geodesics on the de Sitter space. Thus establishing that the kinematic space of the Upper half plane model is the de Sitter space. □

B Gromov's $\delta-$ Hyperbolicity

Graphs are ubiquitous data structures capable of modeling diverse data from disparate fields such as Biology (Protein-Protein interaction), Chemistry (molecules), Social and Internet Networks. Recent trends in machine learning on graph data includes embedding graph nodes in hyperbolic space, with the intent of preserving the hierarchical relationship between entities in the data. Mathematically, we characterize the *embeddability* of a hierarchical data in hyperbolic space by a concept called *hyperbolicity*, proposed by Gromov [16]. Let $\{a, b, c, d\}$ be the vertices of the graph $G(V, E)$ and let ($\mathbb{S} = \{S_1 = d(a, b) + d(d, c)\}$, $\{S_2 = d(a, c), d(b, d)\}$, and $\{S_3 = d(a, d) + d(b, c)\}$). The $\delta(a, b, c, d)$ is given by

$$\delta(a, b, c, d) = \frac{1}{2} \max_{\{a,b,c,d\} \in V(G)} \text{hyp}(a, b, c, d) \tag{23}$$

where, hyp(a, b, c, d) = Difference of two largest values in $\mathbb{S}$. The graph $G(V, E)$ can be viewed as a metric space with $d(.)$ giving distance (geodesic) between vertices. The lower the δ value, the more *Tree-like* a given graph data is, and thus it can be isometrically embedded in hyperbolic space.